How to Kernelize SVM

$$\mathcal{L}_D(\alpha) = \sum_{i=1}^{N} \alpha_i - \frac{1}{2} \sum_{j=1}^{N} \sum_{i=1}^{N} \alpha_i \alpha_j y_i y_j X_i^T X_j$$

$$W = \sum_{i=1}^{N} \alpha_i y_i X_i$$

$$b = -\frac{1}{2}[\max_{i:\, y_i = -1} W^T X_i + \min_{i:\, y_i = +1} W^T X_i]$$

Remember !

Class of vector Z = sign(W^T Z + b)

Mapping Instead of using X_i (of dimension d) use $\phi(X_i)$ *(with higher dimension)*

$$X_i^T X_j \Rightarrow \phi(X_i)^T \phi(X_j)$$

Kernelization Instead of using $\varphi(X_i)^T \varphi(X_i)^T$ use $K(X_i^T, X_j)$ *(Where K is one of kernel functions)*

$$\mathcal{L}_D(\alpha) = \sum_{i=1}^{N} \alpha_i - \frac{1}{2} \sum_{j=1}^{N} \sum_{i=1}^{N} \alpha_i \alpha_j y_i y_j (X_i^T X_j)^2$$

$$W = \sum_{i=1}^{N} \alpha_i y_i X_i$$

$$b = -\frac{1}{2}[\max_{i:\, y_i = -1} [W^T X_i]^2 + \min_{i:\, y_i = +1} [W^T X_i]^2]$$

Class of vector Z is sign([W^T Z]^2 + b)